Engine Model(s): PT6A

72-30-04

GAS GENERATOR CASE - DESCRIPTION AND OPERATION

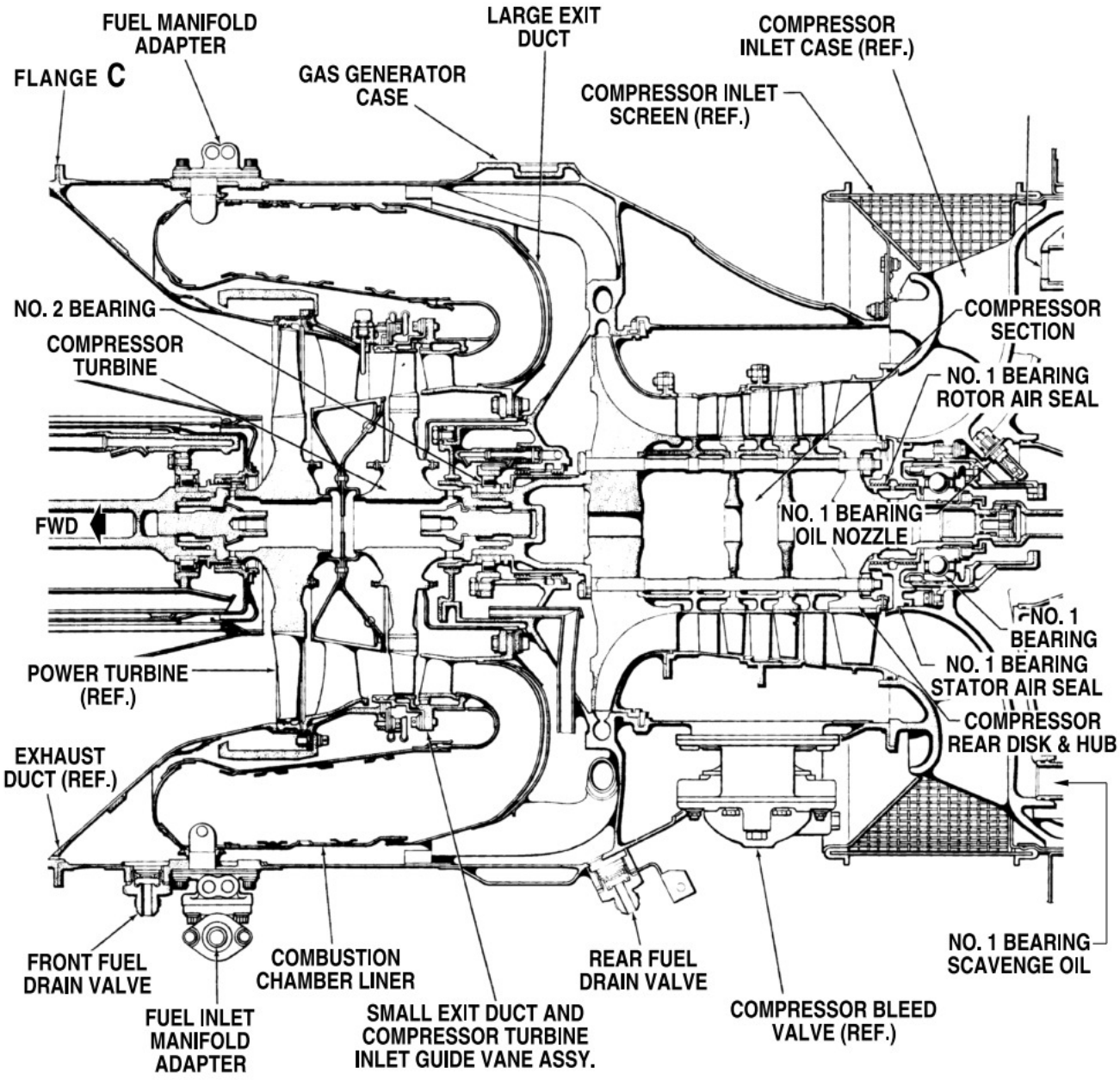
1. General

The gas generator case comprises the components necessary for the compression of the incoming air, and the mixing and combustion of fuel and air. It also provides the main attachment points for mounting the engine in the airframe.

2. Description and Operation (Ref. Fig. 1)

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

Figure 1 Gas Generator Case - Cross-section



c7926a

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement

The gas generator case is located between the compressor inlet case at the rear and the exhaust duct at the front. The case consists of two stainless steel sections fabricated into a single structure, and may be coated with a diffused aluminide process treatment for increased resistance to corrosion attack. The rear inner section provides a housing for the compressor rotor and stator assembly. A bleed port for the compressor bleed valve (Ref. 75-30-00) is located at the 7 o'clock position on the conical outer section of the case. The No. 2 bearing, with compressor and compressor turbine stator air seals, is located in the case centerbore. The compressor double labyrinth stator air seal is secured in the centerbore by a double coil retaining ring, while the compressor turbine stator air seal is secured to the front flange at the centerbore by eight bolts.

Diffuser tubes, brazed inside the center section of the case, increase the static pressure of the compressor discharge air and direct it through straightening vanes to the combustion area. Compressor discharge air (P3) is piped from a diffuser tube at approximately the 4 o'clock position to a boss at the 5 o'clock position on the case to provide controlling pressure for the fuel control system (Ref. 73-20-00). A vent pipe located at the 8 o'clock position on the case provides P3 air for bleed valve operation.

Pressure oil to lubricate the No. 2 bearing is applied to the bearing area from a inlet port at the 5 o'clock position on the gas generator case, an internal pressure tube and a cored passage in the centerbore of the case. A dual nozzle assembly at the 2 o'clock position on the cored passage applies oil to the front and rear faces of the bearing. Scavenge oil from the bearing area is returned to the accessory gearbox through an internal tube installed at the 6 o'clock position in the bearing area to a boss at the 5 o'clock position on the gas generator case. An external transfer tube connects the boss to a similar boss on the accessory gearbox.

The front section of the gas generator case forms the outer housing for the combustion chamber liner. The case is of cylindrical form, with mounting bosses for the 14 fuel manifold adapters, the front and rear drain valves and the two glow plugs or spark igniters. The drain valves are located at the 6 o'clock position while the glow plugs or spark igniter bosses are at the 4 and 9 o'clock positions.

The glow plugs or spark igniters and the fuel nozzles and sheaths of the manifold adapters pass through the case and outer wrapper of the combustion chamber liner to protrude into the combustion area. The fuel nozzle sheaths are utilized to support the front end of the combustion chamber liner. Three pads are provided on the gas generator case at the 4, 8 and 12 o'clock positions for engine mounting in the airframe.

Printed on 10/NOV/26 P&WC Proprietary – subject to restrictions in Technical Data Agreement